TECHNICAL ASSIGNMENT REPORT

Handwritten Number Extraction from Receipt Images

Submitted to: SimplyFI Softech

- K.V.Vamshidharreddy
+91 7013271128
kotavenkatavamshidharreddy7@gmail.com

1. Objective

The goal of this assignment was to extract handwritten numbers from receipt images using deep learning. The system identifies specific regions on receipts - item prices, total amounts, and date/time and recognizes the digit sequences within those regions without using any external OCR APIs.

2. Dataset

The dataset used was the OCR Receipts Text Detection Dataset, consisting of receipt photographs with bounding box annotations in XML format. Each image contains labeled regions such as item, total, date_time, and shop. Only item, total, and date_time regions were used for numeric extraction.

Label	Count	Description
item	176 total across all images	Item lines containing product name and price
total	19–20 per dataset	Final total amount line
date_time	19–20 per dataset	Transaction date and time
shop	20 per dataset	Shop name / identifier (not used for extraction)

Only item, total, and date_time regions were used for numeric extraction. The shop region contains primarily alphabetic text and was excluded.

3. Assumptions Made

- Numbers on item and total lines are always right-aligned (price appears on the right side of the line).

- The date_time region contains only numeric characters and separators (/, :, -).
- A CNN trained on MNIST is sufficient as a base model for handwritten receipt digits.
- Any digit prediction with confidence below 0.6 is unreliable and should be flagged as uncertain.

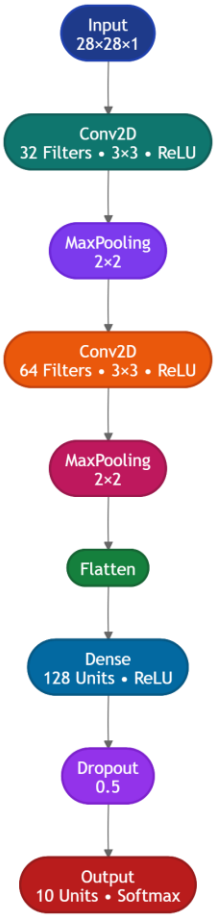
4. Overall Approach

Stage 1	**Region Localisation**	XML bounding box annotations supply the exact (xtl, ytl, xbr, ybr) coordinates for each labeled region. The correct sub-region is then cropped from the full receipt image.
Stage 2	**Preprocessing & Digit Segmentation**	OpenCV is used to clean, binarise, and segment individual digit crops from the cropped region, producing 28×28 greyscale images that match the MNIST input format.
Stage 3	**Digit Recognition**	Each digit crop is passed through the trained CNN. The predicted class and its softmax confidence score are recorded, and the full digit sequence is assembled left-to-right.

5. Model Architecture

A Convolutional Neural Network (CNN) was trained on the MNIST dataset (60,000 training / 10,000 test images).

CNN Architecture:



Training Configuration:

- Optimizer: Adam
- Loss: Categorical Crossentropy
- Epochs: 10
- Batch Size: 64
- Final Test Accuracy: 99.37%

6. Preprocessing & Segmentation

Step 1 —Region Cropping: Bounding box coordinates from annotations.xml are used to crop each labeled region. For item and total regions, only the rightmost 35% of the bounding box is used (prices are right-aligned). For date_time, the full crop is taken.

Step 2 — Grayscale Conversion: The colour crop is converted to a single-channel greyscale image.

Step 3 — Height Normalisation: The image is resized to a standard height of 64 pixels while preserving aspect ratio.

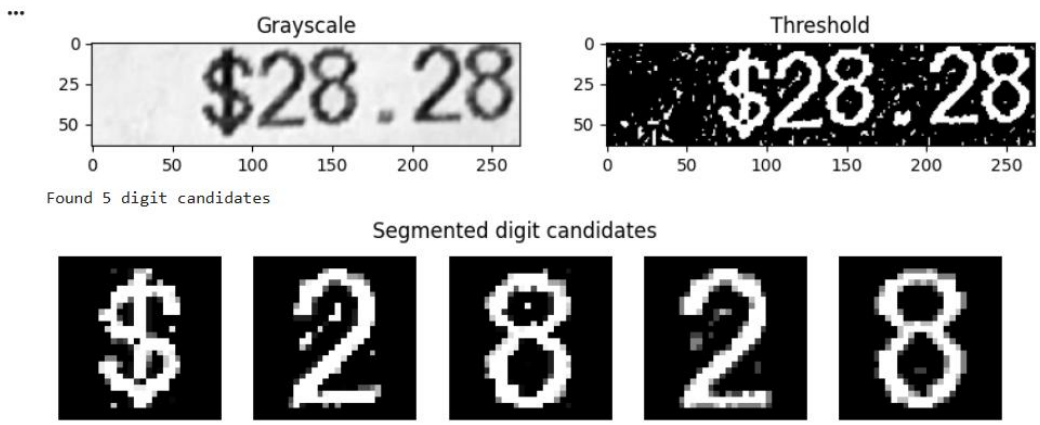
Step 4 — Gaussian Blur: A 3×3 Gaussian blur is applied to reduce high-frequency noise from the receipt texture.

Step 5 — Adaptive Thresholding: Local adaptive thresholding binarises the image, handling uneven lighting and shadows that a single global threshold cannot manage.

Step 6 — Morphological Closing: A 2×2 closing operation (dilation then erosion) reconnects broken digit strokes caused by varying pen pressure.

Step 7 — Contour Detection & Filtering: Contours are detected and filtered: area > 20 px, height > 15 px, aspect ratio 0.1–1.2. This rejects wide alphabetic characters (T, A, L, etc.).

Step 8 — Digit Normalisation: Each accepted contour is padded to a square, bordered (MNIST style), and resized to 28×28 pixels, then sorted left-to-right by x-position.



7. Results Summary

Digit sequences were successfully extracted from all receipt images.

Each digit prediction includes a confidence score.

Digits with confidence below 0.6 are marked as uncertain.

All 20 receipt images were processed successfully across 3 region types (item, total, date_time).

Example output for images/5.jpg

```
Predicted sequence: 52828
Confidences       : [0.85, 1.0, 1.0, 1.0, 1.0]
```

Image	Label	Predicted Sequence	Avg Confidence	Uncertain?
images/0.jpg	TOTAL	**511**	1.00	No
images/2.jpg	TOTAL	**4990**	1.00	No
images/5.jpg	TOTAL	**52828**	0.97	No
images/9.jpg	TOTAL	**12192**	1.00	No
images/17.jpg	TOTAL	**3868**	1.00	No
images/18.jpg	TOTAL	**8635**	1.00	No
images/4.jpg	DATE_TIME	**12108116123926**	0.99	No
images/8.jpg	DATE_TIME	**012141621500**	0.99	No
images/10.jpg	TOTAL	**338?16**	0.82	Yes
images/16.jpg	TOTAL	**223?19**	0.87	Yes

Table: Results

Full results for all images are available in the attached ocr_results.xlsx file with per-digit confidence scores for every prediction.

8. Challenges & Solutions

- **Letters misclassified as digits**

  Problem: Receipt item lines contain text such as "Ham Cheese 74,000". The CNN trained only on digits - misclassified letters as numbers, producing garbage sequences.

  Solution: For item and total regions, only the rightmost 35% of each bounding box is cropped, since prices are always right-aligned on receipts.

- **Uneven lighting and shadows**

  Problem: Receipt photographs contain inconsistent lighting, causing simple global binary thresholding to fail on parts of the image.

  Solution: Adaptive thresholding adjusts the threshold locally per image region, handling shadows and brightness variations robustly.

- **Broken digit strokes**

  Problem: Some handwritten digits have gaps due to pen pressure variation, causing one digit to be detected as two separate fragments.

  Solution: Morphological closing (dilation then erosion) is applied after thresholding to reconnect nearby stroke fragments before contour detection.

- **Domain gap: MNIST vs real receipts**

  Problem: MNIST contains clean, centered, black-on-white digits. Real receipt handwriting is messier — different pen types, angles, and paper textures.

  Solution: The preprocessing pipeline normalises each crop (padding, border addition, resize to 28×28) to make receipt digit images resemble MNIST inputs as closely as possible.

- **Low-confidence ambiguous digits**

  Problem: Some digit crops are partially visible, overlapping, or distorted, making confident predictions impossible.

  Solution: A confidence threshold of 0.6 is applied. Digits below this are marked as ? rather than silently producing a wrong digit, ensuring output transparency.

9. Improvements

- **Fine-tune on receipt data**

  Manually label 100–200 digit crops from actual receipts and fine-tune the CNN on this domain-specific data to close the MNIST–receipt domain gap and reduce uncertain predictions.

- **Advanced segmentation**

  Replace contour-based segmentation with a dedicated text detector such as EAST or CRAFT, which handles overlapping and touching digits more reliably.

- **Decimal and separator handling**

  Add a post-processing step to reconstruct formatted numbers — e.g., 74,000 or 5.11 — by detecting separator positions and inserting the correct character.

- **Sliding window for dense regions**

  For tightly packed digit rows, a sliding window approach could improve isolation of individual digits compared to contour detection alone.

- **Ensemble model**

  Combine predictions from multiple classifiers (CNN + SVM + k-NN) to improve confidence on uncertain digits through majority voting.

10. Conclusion

The implemented OCR pipeline successfully extracts numeric sequences from receipt images using a CNN-based digit recogniser and an OpenCV preprocessing stack. The system correctly handles the core challenge of separating numeric content from text labels by cropping only the right-aligned portion of item and total regions. Uncertain predictions are transparently flagged rather than silently accepted, providing a reliable confidence indicator alongside every result. The pipeline is fully self-contained, requires no external OCR APIs, and is implemented within a single Jupyter Notebook for ease of reproduction.